



Thermodynamics & Thermochemistry

Grade 11 • Chemistry • Session 3





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Yes

No

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Teacher Introduction

Taught and Mentored
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Produced bulk students in
GMCs and IITs

ME
BITS Pilani

7+ years of teaching
experience

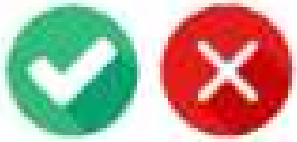
GOVIND MANGLANI

CHEMISTRY EXPERT

Chat Off

MUTUAL AGREEMENT

- Hi everyone! Welcome to the AI Live Class!
- This is a recorded interactive Session, but all polls, quizzes, hotspots, etc are live and we will provide personalized feedback to you.
- Your complete participation would be evaluated for the leaderboard.
- There are highly qualified Class Teachers in the class so please be assured that all your doubts will be answered.
- I know some concepts are tricky and hence will revise every concept after its completion.
- I request you to be well behaved and not spam the public chat.
- Try to make the best use of this class and enjoy the session with your classmates :)



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Recap.



Recap.

Zeroth Law of Thermodynamics

If two systems are in thermal equilibrium with third system, then they are also in thermal equilibrium with each other.

First Law of Thermodynamics

Energy can neither be created nor be destroyed.

$\Delta U = q + w$

Quantity	Positive Sign (+)	Negative Sign (–)
For q	system gains heat	system loses heat
w	work done on system	work done by system
ΔU	net gain of energy by system	net loss of energy by system

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Let's Discuss

What is First Law of Thermodynamics ?



Learning Outcomes

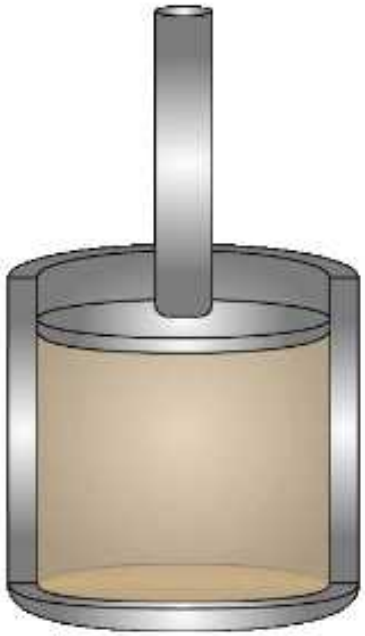
1. Calculation of Work
2. Applications of First Law of Thermodynamics
3. Heat Capacity

Learning Outcomes

1. **Calculation of Work**
2. Applications of First Law of Thermodynamics
3. Heat Capacity

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Calculation of Work (w)





Compression

$\triangle V$ = –ve

W = +ve

Work is done on the system

Expansion

$\triangle V$ = +ve

W = –ve

Work done by the system

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Calculation of Work (w)

$$W_{rev} = -\int_{V_i}^{V_f} p_{ex}\, dV = -\int_{V_i}^{V_f} (p_{in} \pm dp)\, dV$$

Since $dp \times dV$ is very small we can write

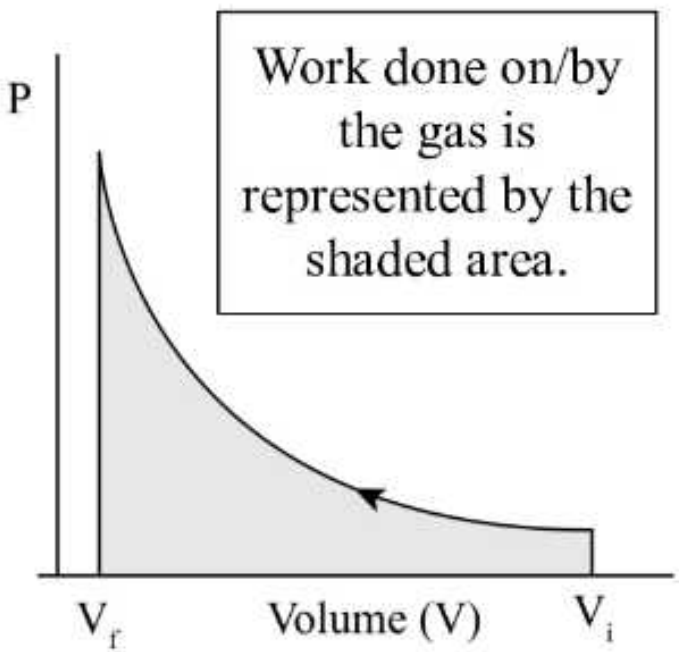
$$w_{rev} = -\int_{V_i}^{V_f} p_{in}\, dV$$

For 'n' mol of an ideal gas, $pV = nRT$

$$\Rightarrow p = \frac{nRT}{V}$$

Therefore, work done can also be calculated as,

$$w_{rev} = -\int_{V_i}^{V_f} nRT\, \frac{dV}{V}$$



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Calculation of Work (w)

Work in Isothermal process

Work in reversible process

$$W_{PV} = -\int_{V_1}^{V_2} P_{ext}\,dV$$

$$P_{gas} = P_{ext} + dP$$

$$P_{gas} \cong P_{ext}$$

$$W_{rev} = -\int_{V_1}^{V_2} P_{gas}\,dV$$

$$W_{rev} = -nRT \ln\left(\frac{V_2}{V_1}\right)$$

Work in irreversible process

$$W_{PV} = -\int_{V_1}^{V_2} P_{ext}\,dV$$

$$P_{ext} = \text{constant}$$

$$W_{irr} = -P_{ext}(V_2 - V_1)$$

Calculation of Work (w)

Work in Isochoric process

V = constant

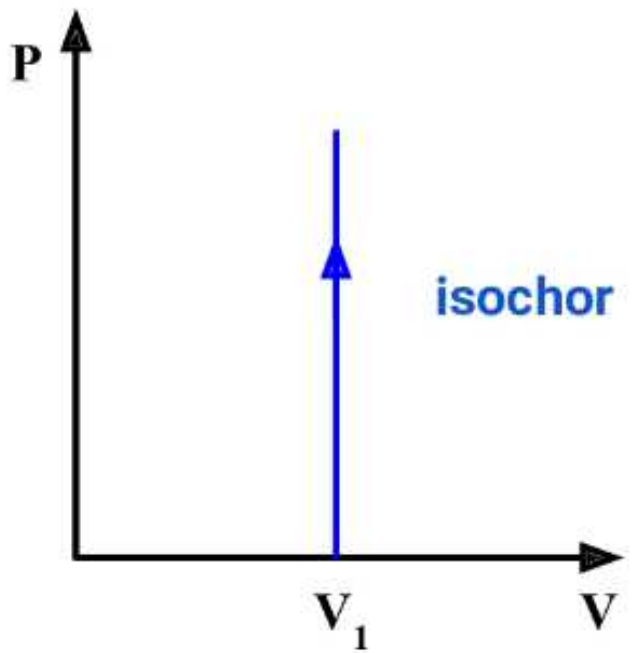
dV = 0

W = 0

From first law of thermodynamics

$\Delta U = q_v$

Heat exchanged at constant volume



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Calculation of Work (w)

Work in Isobaric process

P_{gas} = constant

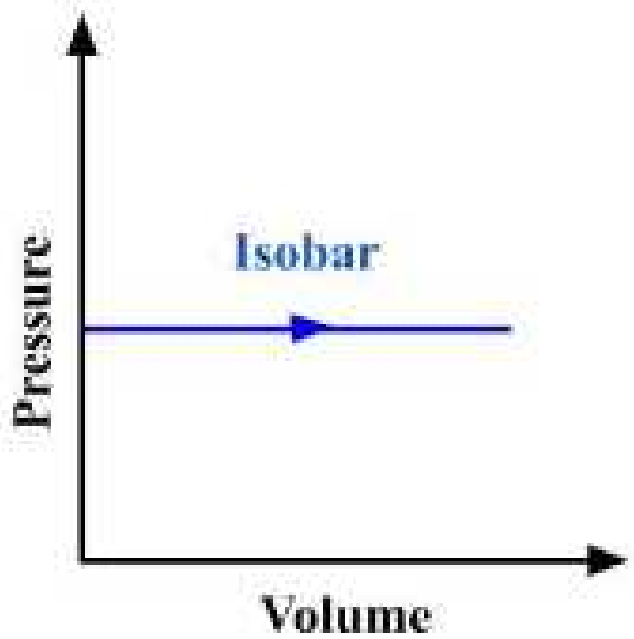
Since pressure of gas is defined during whole of the process and hence the process must be reversible

So $dw = -P_{ext} dV$

$= -P_{gas} dV$

$= -P_{gas} \int_{V_i}^{V_f} dV$

$W = -P(V_f - V_i)$



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Interactivity Pod

For which thermodynamic process work done is zero?

A Isobaric process

B Isochoric process

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Interactivity Pod

For which thermodynamic process work done is zero?

A Isobaric process

B Isochoric process

For isochoric process,

V = constant

$dV = 0$

$W = -PdV = 0$

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Let's Solve.



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Let's Solve.

What is the work done by 1.00 mol of an ideal gas expanding reversibly from a volume of 22.4 L to a volume of 44.8 L at a constant temperature of 273 K?

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Let's Solve.

What is the work done by 1.00 mol of an ideal gas expanding reversibly from a volume of 22.4 L to a volume of 44.8 L at a constant temperature of 273 K?

Solution:

$$w = -nRT \ln\left(\frac{V_2}{V_1}\right)$$

$$w = -(1.00 \cancel{\text{mol}})\left(8.314 \frac{\text{J}}{\cancel{\text{mol}}\,\cancel{\text{K}}}\right)(273 \cancel{\text{K}}) \ln\left(\frac{44.8\ \text{L}}{22.4\ \text{L}}\right)$$

$$= -1570\ \text{J} = -1.57\ \text{kJ}$$

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Doubts Time



I will now be taking frequently asked doubts

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Doubts



- Why work done in isothermal reversible process and isothermal irreversible process is not same ?

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Quiz.





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Quiz.

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What is the work done when a gas contracts irreversibly from 377 mL to 177 mL under a constant pressure of 1520 torr while at the same time being cooled by removing 124 J of heat? (Take 1 atm = 760 torr and 1 atm L = 100 J)

A. 10 J

B. 20 J

C. 30 J

D. 40 J

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Quiz.

Chat On

What is the work done when a gas contracts irreversibly from 377 mL to 177 mL under a constant pressure of 1520 torr while at the same time being cooled by removing 124 J of heat? (Take 1 atm = 760 torr and 1 atm L = 100 J)

A 10 J

B 20 J

C 30 J

D 40 J

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Quiz.

Chat On

What is the work done when a gas contracts irreversibly from 377 mL to 177 mL under a constant pressure of 1520 torr while at the same time being cooled by removing 124 J of heat? (Take 1 atm = 760 torr and 1 atm L = 100 J)

Solution

The given process is irreversible. Heat is removed from the system, so its value will be negative, $q = -124$ J.

$$P_{ext} = 1520 \text{ torr} = \frac{1520}{760} \text{atm} = 2 \text{ atm}$$

$$W = -P_{ext}(V_2 - V_1) = -2(0.177 - 0.377)$$
$$W = 0.4 \text{ atm L}$$
$$W = 0.4 \times 100 \text{ J} = 40 \text{ J}$$

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Quiz.

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A real gas is subjected to an adiabatic process from (2 bar, 40 L, 300 K) to (4 bar, 30 L, 300 K) against a constant pressure of 4 bar. Calculate the work done for the process. (Take 1 bar L = 100 J)

A. 4000 J

B. 3432 J

C. 40 J

D. 6321 J

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Quiz.

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A real gas is subjected to an adiabatic process from (2 bar, 40 L, 300 K) to (4 bar, 30 L, 300 K) against a constant pressure of 4 bar. Calculate the work done for the process. (Take 1 bar L = 100 J)

A. 4000 J

B. 3432 J

C. 40 J

D. 6321 J

Solution

Work done is given by:

$W = -P_{ext}(V_2 - V_1)$

$W = -4(30 - 40)$

$W = 40 \text{ bar L} = 40 \times 100 = 4000 \text{ J}$

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Milestone Achieved

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Learning Outcomes

1. Calculation of Work
2. **Applications of First Law of Thermodynamics**
3. Heat Capacity

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Let's Discuss

How work is calculated in irreversible process ?

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Applications of First Law of Thermodynamics

1) Isothermal Process

When a system undergoes an isothermal change, $\Delta U = 0$

Then, according to First law of thermodynamics,

$\mathbf{\Delta U = q + w \Rightarrow q = -w}$

Work done in isothermal irreversible process:

$w = -p_{ex} \Delta V$

Work done in isothermal reversible process:

$$w = -nRT \int \frac{dV}{V} = -2.303nRT \log \frac{V_2}{V_1} = -2.303nRT \log \frac{P_1}{P_2}$$

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Applications of First Law of Thermodynamics

2) Isobaric Process

For isobaric process, the work done can be calculated as $-\int p.dV$.

Then, according to First law of thermodynamics

$$\Delta U = q - \int p.dV$$

Free expansion: Expansion of a gas in vacuum ($p_{ex} = 0$) is called free expansion. No work is done during free expansion of an ideal gas.

Applications of First Law of Thermodynamics

3) Isochoric Process

For isochoric process, the work done = 0

Then, according to First law of thermodynamics

$\mathbf{\Delta U = q}$

Thus, internal energy change represents the heat change taking place at constant volume. $\Delta U = q_v$

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Applications of First Law of Thermodynamics

4) Cyclic Process

In a cyclic process, where initial and final states are the same,

$U_2 = U_1$, then $\Delta U = 0$

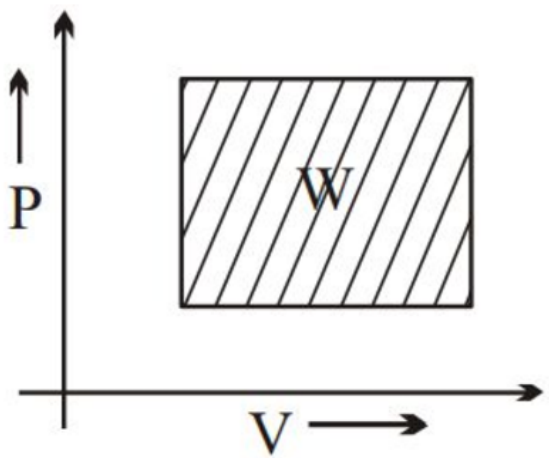
i.e, $\mathbf{q = -w}$

Calculation of work for cyclic process:

Work done = Area enclosed in PV-diagram

For clockwise it is –ve.

For anti-clockwise it is +ve.



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Interactivity Pod

Work done during a cyclic process is zero. The statement is

A True

B False

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Interactivity Pod

Work done during a cyclic process is zero. The statement is

A True

B False

For a cyclic process, the initial and final states are same,

$U_2 = U_1$, then $\Delta U = 0$

$q = -w$

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Let's Solve.



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Let's Solve.

Calculate the work performed (in cal.) when 2 moles of hydrogen expand isothermally and reversibly at 25°C from 15 to 50 litres.
(log 3.3 = 0.52)

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Let's Solve.

Calculate the work performed (in cal.) when 2 moles of hydrogen expand isothermally and reversibly at 25°C from 15 to 50 litres. (log 3.3 = 0.52)

Solution

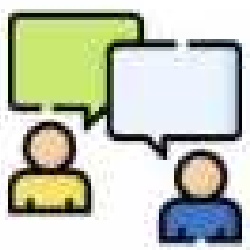
We have, $W = -2.303\ n\ RT \log \frac{V_2}{V_1} = -2.303 \times 2 \times 2 \times 298 \times \log \frac{50}{15}$
$= -1427.5$ calories.

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Silent Mode

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No work is done during free expansion of an ideal gas .(T/F)



Discuss among yourselves

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Doubts Time



I will now be taking frequently asked doubts

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Doubts



- Why the work done in a cyclic process is taken as –ve if area enclosed in PV-diagram is clockwise ?

Quiz.





Quiz.

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Chat Off

Which one of the following equations does not correctly represent the first law of thermodynamics for the given processes involving an ideal gas? (Assume non-expansion work is zero)

A. Cyclic process : $q = -w$

B. Adiabatic process : $\Delta U = -w$

C. Isochoric process : $\Delta U = q$

D. Isothermal process : $q = -w$

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Quiz.

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Which one of the following equations does not correctly represent the first law of thermodynamics for the given processes involving an ideal gas? (Assume non-expansion work is zero)

A. Cyclic process : $q = -w$

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D. Isothermal process : $q = -w$

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Quiz.

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Which one of the following equations does not correctly represent the first law of thermodynamics for the given processes involving an ideal gas? (Assume non-expansion work is zero)

Solution

Form first law of thermodynamics, $\Delta U = q + w$

For adiabatic process, $q = 0$

$\therefore \quad \Delta U = w$

For isothermal process, $\Delta U = 0 \Rightarrow q = -w$

For cyclic process, $\Delta U = 0 \Rightarrow q = -w$

For isochoric process, $w = 0 \Rightarrow \Delta U = q$

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Quiz.

Chat Off

$1\ J = 10^7$ ergs

1 L-atm = 101.33

1 L-bar = 100 J

The work done in ergs for the reversible expansion of one mole of an ideal gas from a volume of 10 litres to 20 litres at 25^0C is

→ 298K

A $-2.303 \times 298 \times 0.082 \log 2$

B $-298 \times 10^7 \times 8.31 \times 2.303 \log 2$

C $-2.303 \times 298 \times 0.082 \log 0.5$

D $-8.31 \times 10^7 \times 298 \times 2.303 \log 0.5$

$W = -2.303 nRT \log \frac{V_f}{V_i}$

$= -2.303 \times 8.314 \times 10^7 \times 298K$

$\log \frac{20}{10}$

$R = 8.314\ J\ K^{-1} mol^{-1}$

$= 8.314 \times 10^7$ ergs $K^{-1} mol^{-1}$ =

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Quiz.

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The work done in ergs for the reversible expansion of one mole of an ideal gas from a volume of 10 litres to 20 litres at 25^0C is

A $-2.303 \times 298 \times 0.082 \log 2$

B $-298 \times 10^7 \times 8.31 \times 2.303 \log 2$

C $-2.303 \times 298 \times 0.082 \log 0.5$

D $-8.31 \times 10^7 \times 298 \times 2.303 \log 0.5$

Solution

We have, $W = -2.303\ n\ RT \log \frac{V_2}{V_1}$

$= -2.303 \times 1 \times 8.31 \times 10^7 \times 298 \times \log \frac{20}{10}$ $\quad (1\ J = 10^7\ erg)$

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Milestone Achieved

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Learning Outcomes

1. Calculation of Work
2. Applications of First Law of Thermodynamics
3. **Heat Capacity**

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Let's Discuss

How work is calculated in isothermal reversible process?

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Heat Capacity

- Heat capacity or thermal capacity is a physical property of matter, defined as the amount of heat to be supplied to an object to produce a unit change in its temperature.
- Heat can be calculated if heat capacity is given to us.
- Heat capacity of a substance is not fixed it is dependent on type of process which is being performed on that substance.
- Let's start understanding different types of heat capacities one by one.

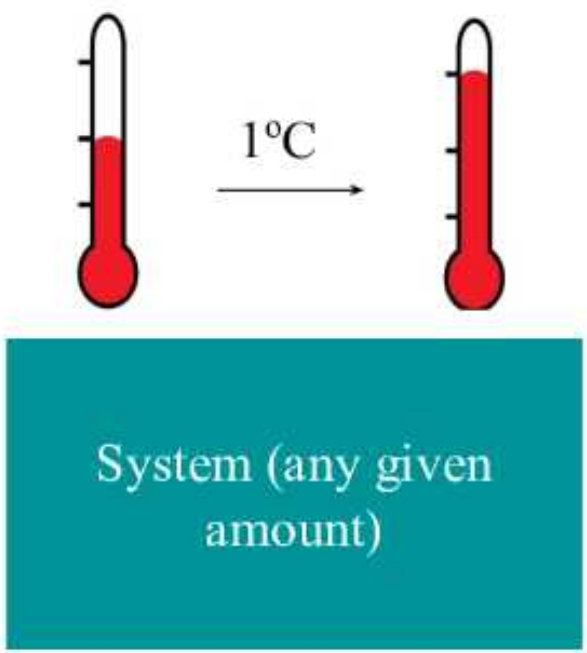
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Heat Capacity:- (c).

Heat Capacity

Total Heat Capacity (C_T)

Heat required to raise the temperature of system by 1°C under the given process is known as **total heat capacity**.



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Heat Capacity

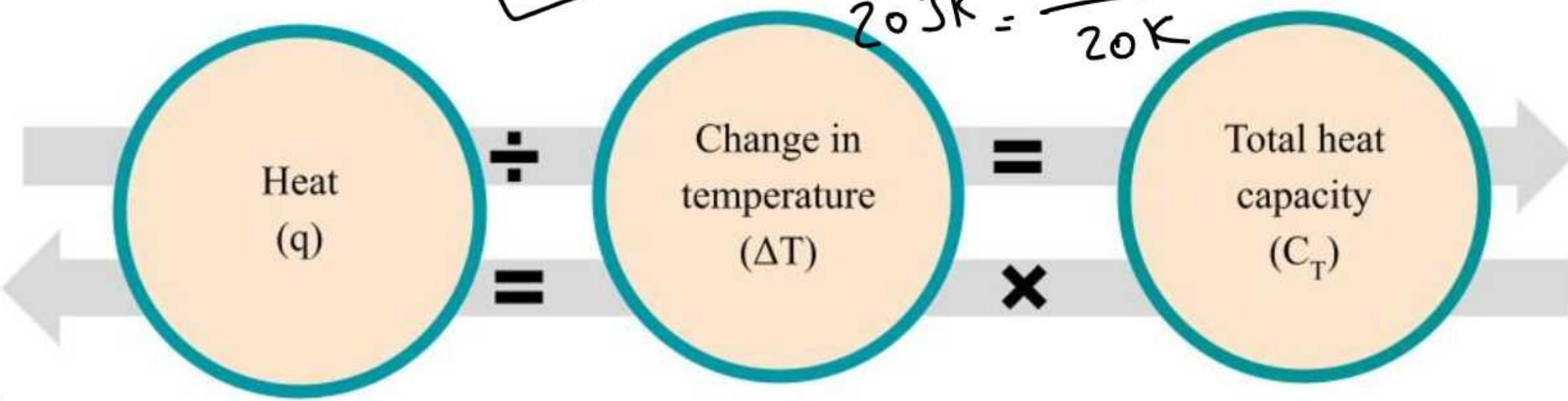
Total Heat Capacity (C_T)

$$C_T = \frac{q}{\Delta T} \quad J\,K^{-1}$$

JK^{-1}

$q = 460\,J$

$$20\,JK^{-1} = \frac{q}{20\,K}$$



$C_T = 20\,JK^{-1}$

200K to 220K.

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Heat Capacity

Total Heat Capacity (C_T)

Mathematically $C_T = \frac{\Delta q}{\Delta T} = \frac{dq}{dT}$ (unit: J/°C)

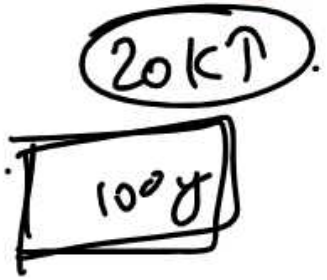
- It is extensive property and path function.

So $dq = C_T dT$

on integrating $q = \int C_T dT$



$C_T = \frac{q}{\Delta T}$



$C_T = \frac{q}{\Delta T}$

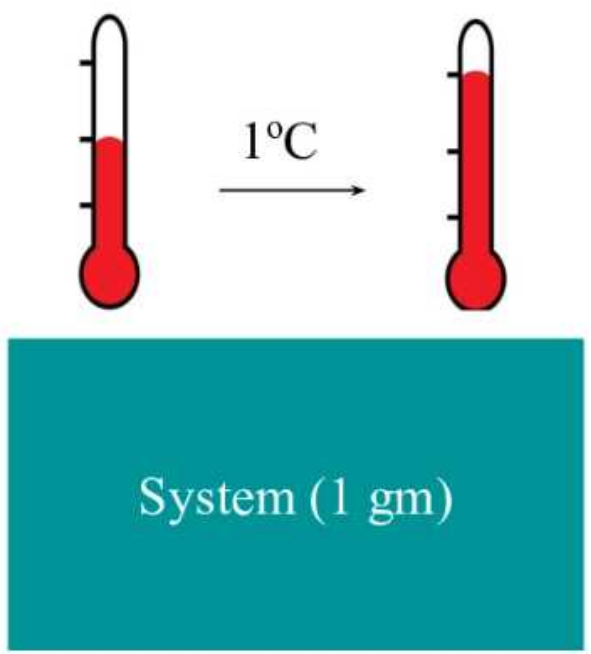
$C_T = \frac{\quad}{20}$ $\quad C_T = \frac{\quad}{20}$

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Heat Capacity

Specific Heat Capacity (s)

Heat required to raise temperature of unit mass (generally 1 g) of a substance by 1 °C.



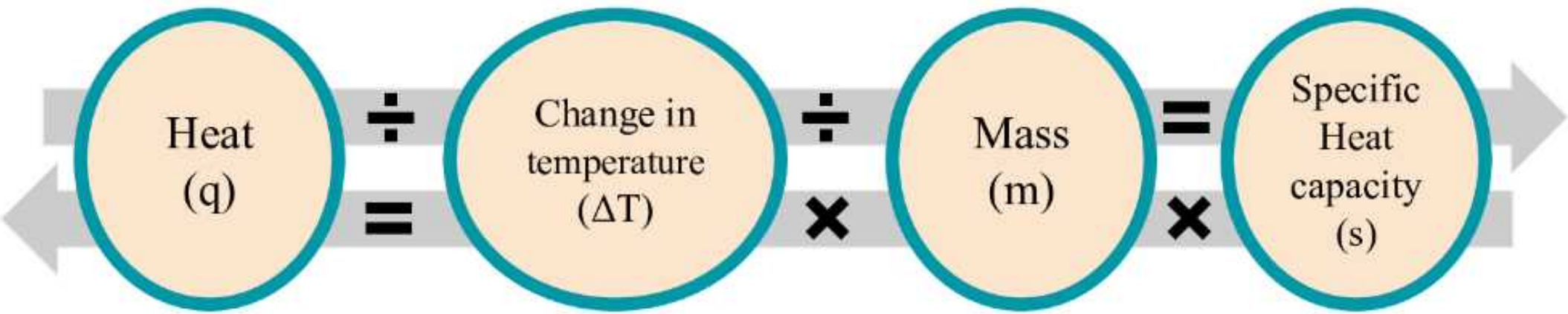
$$s = \frac{q}{m\Delta T}$$

$Jg^{-1}K^{-1}$

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Heat Capacity

Specific Heat Capacity (s)



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Heat Capacity

Specific Heat Capacity (s)

$$s = \frac{\Delta q}{m.\Delta T} = \frac{dq}{m.dT} \text{ (Unit: J } g^{-1} K^{-1})$$

Intensitive property.

So, $dq = m.s.dT$

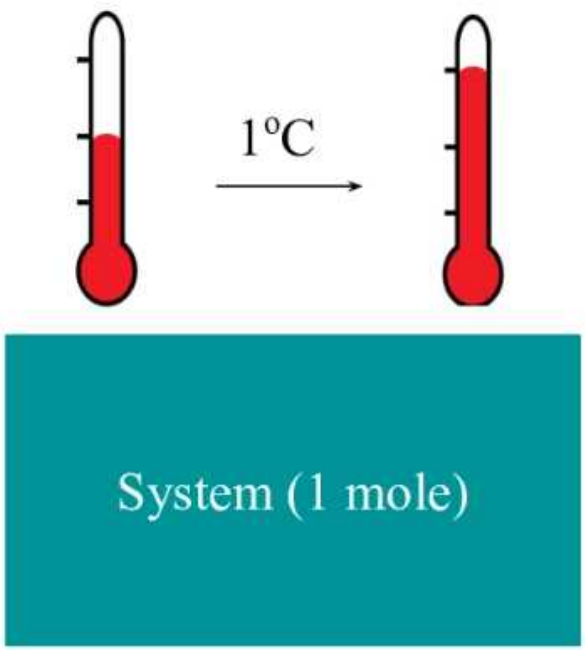
$$q = \int dq = \int m.s.dT = ms\Delta T$$

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Heat Capacity

Molar Heat Capacity (C_m)

Heat required to raise temperature of 1 mole of a substance by 1°C.

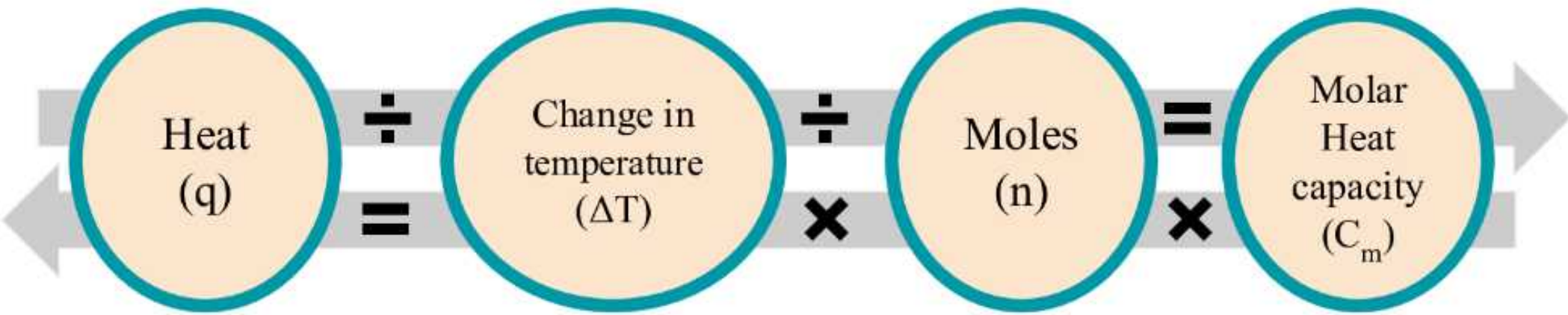


$$C_m = \frac{q}{n\,\Delta T}$$

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Heat Capacity

Molar Heat Capacity (C_m)



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Heat Capacity

Molar Heat Capacity (C_m)

$$C_m = \frac{\Delta q}{n\Delta T} = \frac{dq}{ndT} \text{ (unit = J mole}^{-1}\text{ K}^{-1}\text{)}$$

So $dq = nC_m dT$

$$q = \int nC_m dT = nC_m \Delta T$$

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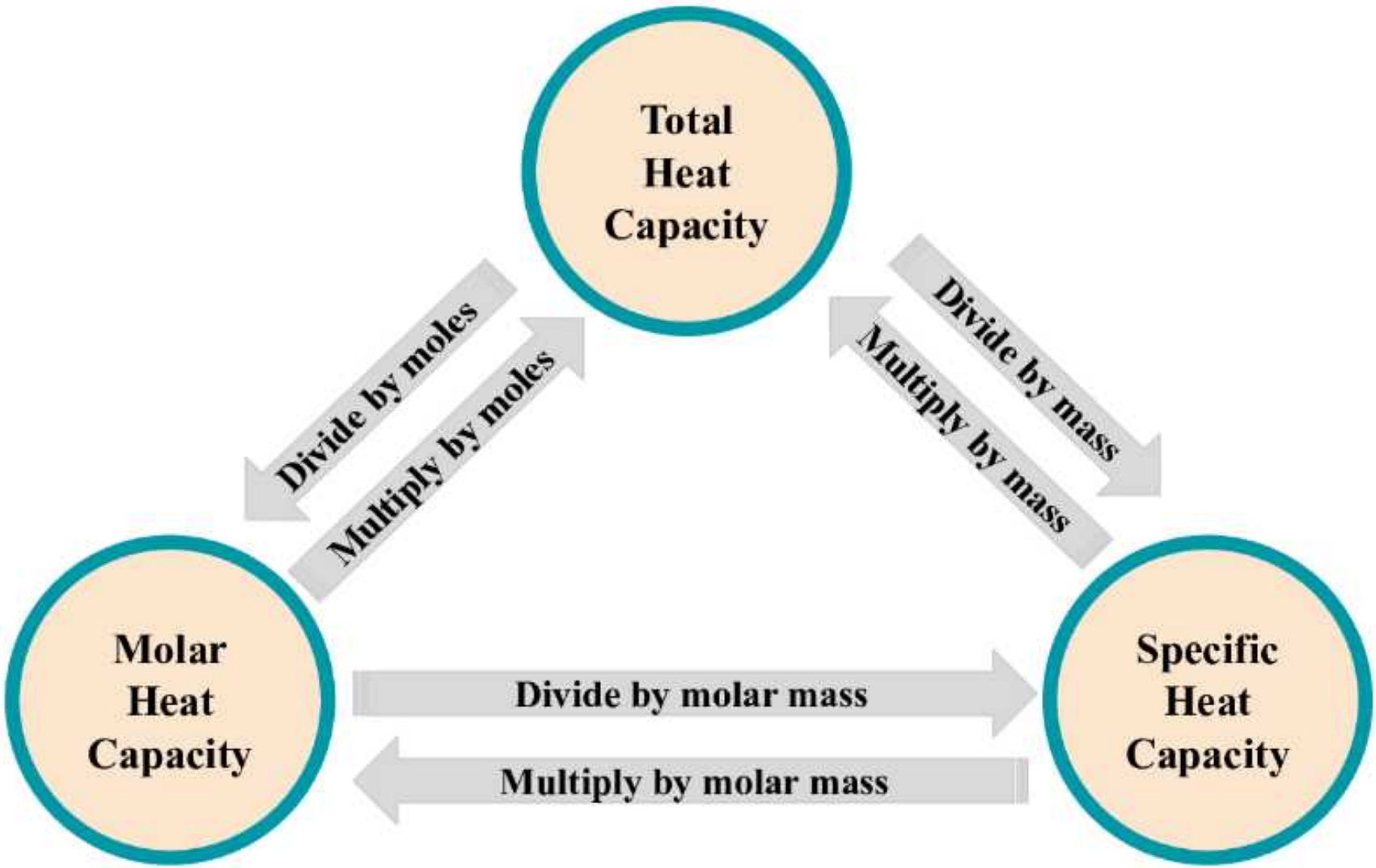
$$C = \frac{q}{\Delta T}$$

$$s = \frac{q}{m \Delta T} = \frac{C}{m} \Rightarrow C = s \times m$$

$$C_m = \frac{q}{n \Delta T} = \frac{C}{n} \Rightarrow C = C_m \times n$$

Heat Capacity

Relation between Heat Capacities



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Chat Off



Interactivity Pod

Heat required to raise the temperature of unit mass of a substance by 1 K is known as

A Molar Heat Capacity

B Specific Heat Capacity

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Interactivity Pod

Heat required to raise the temperature of unit mass of a substance by 1 K is known as

A Molar Heat Capacity

B Specific Heat Capacity

Specific heat capacity is defined as the heat required to raise temperature of unit mass (generally 1 g) of a substance by 1 °C (1K).

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Let's Solve.



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Let's Solve.

209 J is required to increase the temperature of 50.0 g of water by 1.00 K. Thus, the specific heat of water is

$$S = \frac{q}{m \Delta T}$$

$$= \frac{209\ J}{50\ g \times 1\ K}$$

$$= 4.18\ J\ g^{-1}\ K^{-1}$$

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Let's Solve.

209 J is required to increase the temperature of 50.0 g of water by 1.00 K. Thus, the specific heat of water is

Solution

$$s = \frac{q}{m \times \Delta T}$$

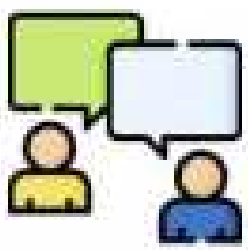
$$s = \frac{209\ J}{(50.0\ g)\ (1.00\ K)} = 4.18\ J/g\text{-}K$$

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Silent Mode

Chat On

Heat required to raise the temperature of system by 1°C under the given process is known as total heat capacity. (T/F)



Discuss among yourselves

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Doubts Time



I will now be taking frequently asked doubts

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Doubts



- For the calculation of heat capacity, temperature in which unit should be used, in °C or K ?
- Which of the type of heat capacities has maximum value for a system ?

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Quiz.



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Quiz.

JEE Mains 2002

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The heat required to raise the temperature of body by 1 K is called

A. Specific heat

B. Heat capacity

C. Water equivalent

D. None

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Quiz.

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The heat required to raise the temperature of body by 1 K is called

A. Specific heat

B. Heat capacity

C. Water equivalent

D. None

Solution

The required to raise the temperature of body by 1 K is heat capacity. Specific heat is defined for 1 g specifically.

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Quiz.

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Calculate the amount of heat required to raise the temperature of 60.0 g of aluminium from 35°C to 55°C. The molar heat capacity of aluminum is 24 J mol^{-1}K^{-1}.

A 1.067 kJ

B 2.215 kJ

C 2.933 kJ

D 1.867 kJ

$C_m = \frac{q}{n\Delta T} \Rightarrow q = C_m \times n \times \Delta T$

$= 24 \times \frac{60}{27} \times (20)$

$=$

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Calculate the amount of heat required to raise the temperature of 60.0 g of aluminium from 35°C to 55°C. The molar heat capacity of aluminum is 24 J $mol^{-1}K^{-1}$.

A 1.067 kJ

B 2.215 kJ

C 2.933 kJ

D 1.867 kJ

Solution

Moles of aluminium(n) = Mass/Molar mass = 60/27

Now, $q = nC_m\Delta T = \frac{60}{27} \times 24 \times (55 - 35) = 1067\ J = 1.067\ kJ$

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Recap.

Total Heat Capacity (C_T)

Heat required to raise the temperature of system by 1°C under the given process is known as total heat capacity.

Specific Heat Capacity (s)

Heat required to raise temperature of unit mass (generally 1 g) of a substance by 1°C.

Molar Heat Capacity (C_m)

Heat required to raise temperature of 1 mole of a substance by 1°C.

Learning Outcomes

1. Relationship between Cp & Cv
2. Calculation of Adiabatic Work

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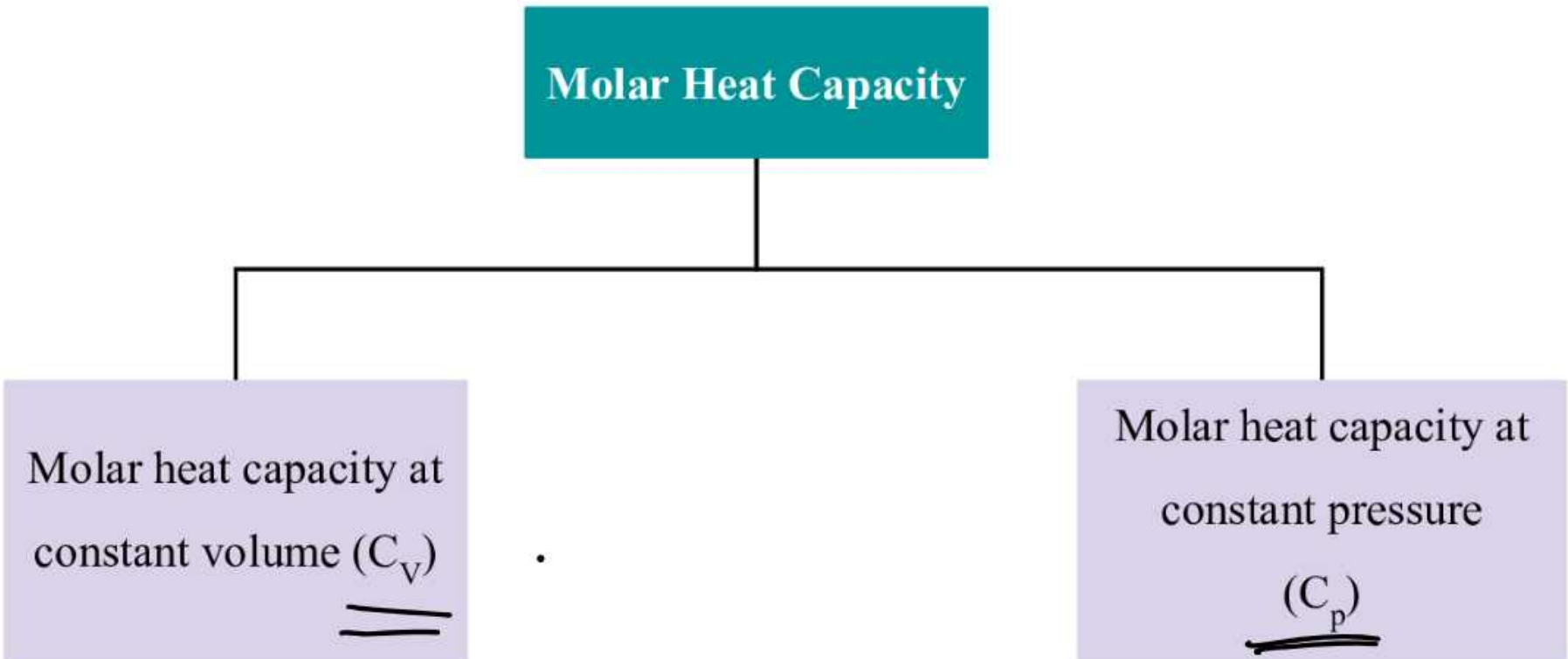
Learning Outcomes

1. Relationship between Cp & Cv
2. Calculation of Adiabatic Work

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Relationship between C_p and C_v

Since heat capacity is a path function. For **an ideal gas**, two different types of molar heat capacities are defined.



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$$C_{m,P} = \frac{q_P}{n \Delta T} \qquad C_{m,V} = \frac{q_V}{n \Delta T}$$

$$q_P = n C_{m,P} \Delta T \qquad q_V = n C_{m,V} \Delta T$$

$$\Delta H = n C_{m,P} \Delta T \qquad \Delta U = n C_{m,V} \Delta T$$

$$q_P = \Delta H$$

$$\Delta U = q + w$$

$$\Delta U = q_V$$

Relationship between C_p and C_v

If the process is isobaric, $q = n C_p \Delta T$

Also, $w = - P\Delta V$ and $\Delta U = n C_V \Delta T$

$\Delta U = q + w$

$\Rightarrow n C_V \Delta T = n C_p \Delta T - (PV_2 - PV_1)$

For an ideal gas, $PV = nRT$

$\Rightarrow n C_V \Delta T = n C_p \Delta T - (nRT_2 - nRT_1)$

$\Rightarrow n C_V \Delta T = n C_p \Delta T - n R \Delta T$

$\Rightarrow C_V = C_p - R$ or $C_p = C_v + R$

$PV = nRT$

$\Delta U = q + w$

$n C_V \Delta T = n C_p \Delta T + P\Delta V$

$n C_V \Delta T = n C_p \Delta T - (PV_2 - PV_1)$

$n C_V \Delta T = n C_p \Delta T - (nRT_2 - nRT_1)$

$n C_V \Delta T = n C_p \Delta T - nR(T_2 - T_1)$

ΔT

$n C_V = n C_p - nR$

$C_p - C_V = R$

$C_V = C_p - R$

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$$C_p - C_v = R$$

Relationship between C_p and C_v

$$C_p = C_v + R$$

- C_p: Molar heat capacity at constant pressure
- C_v: Molar heat capacity at constant volume
- R: Universal gas constant

NOTE: This expression is valid only for an ideal gas.

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$$C_p - C_v = R$$

Relationship between C_p and C_v

Poisson's ratio (γ)

It is the ratio of molar heat capacity at constant pressure and molar heat capacity at constant volume.

$$\gamma = \frac{C_P}{C_V}$$

C_P and C_V in terms of γ and R

Using the relationships, $\gamma = \frac{C_P}{C_V}$ & $C_P = C_V + R$

$$C_V = \frac{R}{\gamma - 1} \qquad C_P = \frac{\gamma R}{\gamma - 1}$$

$C_P - C_V = R$

$\frac{C_P}{C_V} = 1 + \frac{R}{C_V}$

$\gamma = 1 + \frac{R}{C_V}$

$\gamma = \frac{C_V + R}{C_V}$

$C_V = \frac{C_V + R}{\gamma}$

$C_V = \frac{R}{\gamma - 1}$

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$$C_v = \frac{R}{\gamma - 1}$$

$$C_p = \frac{\gamma R}{\gamma - 1}$$

$$C_p = C_v + R$$

$$\gamma = 1 + \frac{R}{C_v}$$

$\therefore$

$$\frac{C_p}{C_v} = 1 + \frac{R}{C_v}$$

$$\gamma - 1 = \frac{R}{C_v}$$

$$C_v = \frac{R}{\gamma - 1}$$

$$C_p = \frac{\gamma R}{\gamma - 1}.$$

$$\frac{C_p}{C_v} = \gamma$$

$$C_p = C_v \cdot \gamma$$

$$= \frac{R}{\gamma - 1} \cdot \gamma$$

$$\boxed{C_p = \frac{\gamma R}{\gamma - 1}}$$

Relationship between C_p and C_v

Table for C_P and C_V for different types of gases

Atomicity	C_V	$C_P = C_V + R$	$\gamma = \frac{C_P}{C_V}$
Monoatomic	$\frac{3}{2}R$	$\frac{5}{2}R$	$\gamma = \frac{5/2R}{3/2R} = 1.66$
Diatomic or linear polyatomic	$\frac{5}{2}R$	$\frac{7}{2}R$	$\gamma = \frac{7/2R}{5/2R} = 1.40$
Non - linear polyatomic	$3R$	$4R$	$\gamma = \frac{4R}{3R} = 1.33$

NOTE: These values are applicable at low or moderate temperatures.

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Interactivity Pod

The correct relationship between C_p and C_v can be represented by

A $C_p + C_v = R$

B $C_p - C_v = R$

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Interactivity Pod

The correct relationship between C_p and C_v can be represented by

A $C_p + C_v = R$

B $C_p - C_v = R$

The correct relationship between Cp and Cv is :

$C_p - C_v = R$

Where R = Universal gas constant

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Let's Solve.



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Let's Solve.

a) $C_p - C_v = R$

b) $\frac{C_p}{C_v} = \gamma$

For a monoatomic ideal gas, the value of C_v is 3 cal mol^{-1} K^{-1}. What will be the value of C_p for the same gas (in cal mol^{-1} K^{-1})?

$C_v = 3$ Cal $mol^{-1} K^{-1}$

$C_p = ?$

$C_p - C_v = R$.

$C_p = 3 + 2$

$C_p = 5$

$R = 8.314\ J K^{-1} mol^{-1}$

$R = 8.314 \times 10^7$ ergs $K^{-1} mol^{-1}$.

$R = 0.0821$ l-atm $K^{-1} mol^{-1}$

$R = 2$ Cal mol^{-1} k^{-1}.

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Let's Solve.

For a monoatomic ideal gas, the value of C_v is 3 cal mol^{-1} K^{-1}. What will be the value of C_p for the same gas (in cal mol^{-1} K^{-1})?

Solution

$$C_p = C_v + R$$
$$C_p = 3 + 2$$
$$C_p = 5 \text{ cal mol}^{-1} \text{ K}^{-1}$$

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Doubts Time



I will now be taking frequently asked doubts

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Doubts



- What is the meaning of atomicity of a gas ?

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Quiz.





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Quiz.

Chat Off

$$C_m = \frac{q}{n \Delta T}$$

The molar heat capacity of water at constant pressure is 75 JK^{-1} mol^{-1}. When 750 J of heat is supplied to 1 mole of water at constant pressure, the increase in temperature of water is

A 10 K

B 0.1 K

C 4.8 K

D 0.24 K

$$q = n C_m \Delta T$$

$$\Delta T = \frac{q}{n C_m} = \frac{750\,J}{1 \times 75} = 10$$

10K.

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The molar heat capacity of water at constant pressure is 75 JK^{-1} mol^{-1}. When 750 J of heat is supplied to 1 mole of water at constant pressure, the increase in temperature of water is

A 10 K

B 0.1 K

C 4.8 K

D 0.24 K

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The molar heat capacity of water at constant pressure is 75 JK^{-1} mol^{-1}. When 750 J of heat is supplied to 1 mole of water at constant pressure, the increase in temperature of water is

Solution

Heat is supplied at constant pressure,

$q_p = nC_p\Delta T$

Thus, $750 = 1 \times 75 \times \Delta T$

$\Delta T = 10\ K$

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Quiz.

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Molar heat capacity(C_P) of D_2O is 10 cal mol^{-1} K^{-1}. How much heat is required to change the temperature of one mole of D_2O by 10 K at constant pressure?

A. 1 cal

B. 100 cal

C. 80 cal

D. None

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Molar heat capacity(C_P) of D_2O is 10 cal mol^{-1} K^{-1}. How much heat is required to change the temperature of one mole of D_2O by 10 K at constant pressure?

A. 1 cal

B. 100 cal

C. 80 cal

D. None

Solution

At constant pressure, $q = n C_P \Delta T$

$q = 1 \times 10 \times 10 = 100$ cal

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Quiz.

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5 moles of an ideal gas at 100 K are allowed to undergo compression till its temperature becomes 200 K. If $C_V = 28\ J\ K^{-1}\ mol^{-1}$, calculate ΔU and ΔpV for this process. ($R = 8.0\ JK^{-1}\ mol^{-1}$)

A. $\Delta U = 14$ kJ; $\Delta(pV) = 18$ kJ

B. $\Delta U = 14$ kJ; $\Delta(pV) = 0.8$ kJ

C. $\Delta U = 14$ kJ; $\Delta(pV) = 4$ kJ

D. $\Delta U = 14$ kJ; $\Delta(pV) = 8.0$ kJ

$q_v = nC_v\Delta T$

$\Delta U = n C_v \Delta T$

$= 5 \times 28 \times 100$

$= 14000\ J$

$= 14\ kJ$

$PV = nRT$

$\Delta(PV) = nR\Delta T$

$= 5 \times 8 \times 100$

$= 4000\ J$

$= 4\ kJ$

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Quiz.

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5 moles of an ideal gas at 100 K are allowed to undergo compression till its temperature becomes 200 K. If $C_V = 28\ J\ K^{-1}\ mol^{-1}$, calculate ΔU and ΔpV for this process. ($R = 8.0\ JK^{-1}\ mol^{-1}$)

A. $\Delta U = 14$ kJ; $\Delta(pV) = 18$ kJ

B. $\Delta U = 14$ kJ; $\Delta(pV) = 0.8$ kJ

C. $\Delta U = 14$ kJ; $\Delta(pV) = 4$ kJ

D. $\Delta U = 14$ kJ; $\Delta(pV) = 8.0$ kJ

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Quiz.

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5 moles of an ideal gas at 100 K are allowed to undergo compression till its temperature becomes 200 K. If $C_V = 28\ J\ K^{-1}\ mol^{-1}$, calculate ΔU and ΔpV for this process. ($R = 8.0\ JK^{-1}\ mol^{-1}$)

Solution

$\Delta U = nC_v\Delta T$

$\Delta U = 5 \times 28 \times (200 - 100) = 14000\ J = 14\ kJ$

Now, for an ideal gas, $\Delta pV = nR\Delta T$

Thus, $\Delta pV = 5 \times 8 \times (200 - 100) = 4000\ J = 4\ kJ$

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Milestone Achieved

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Learning Outcomes

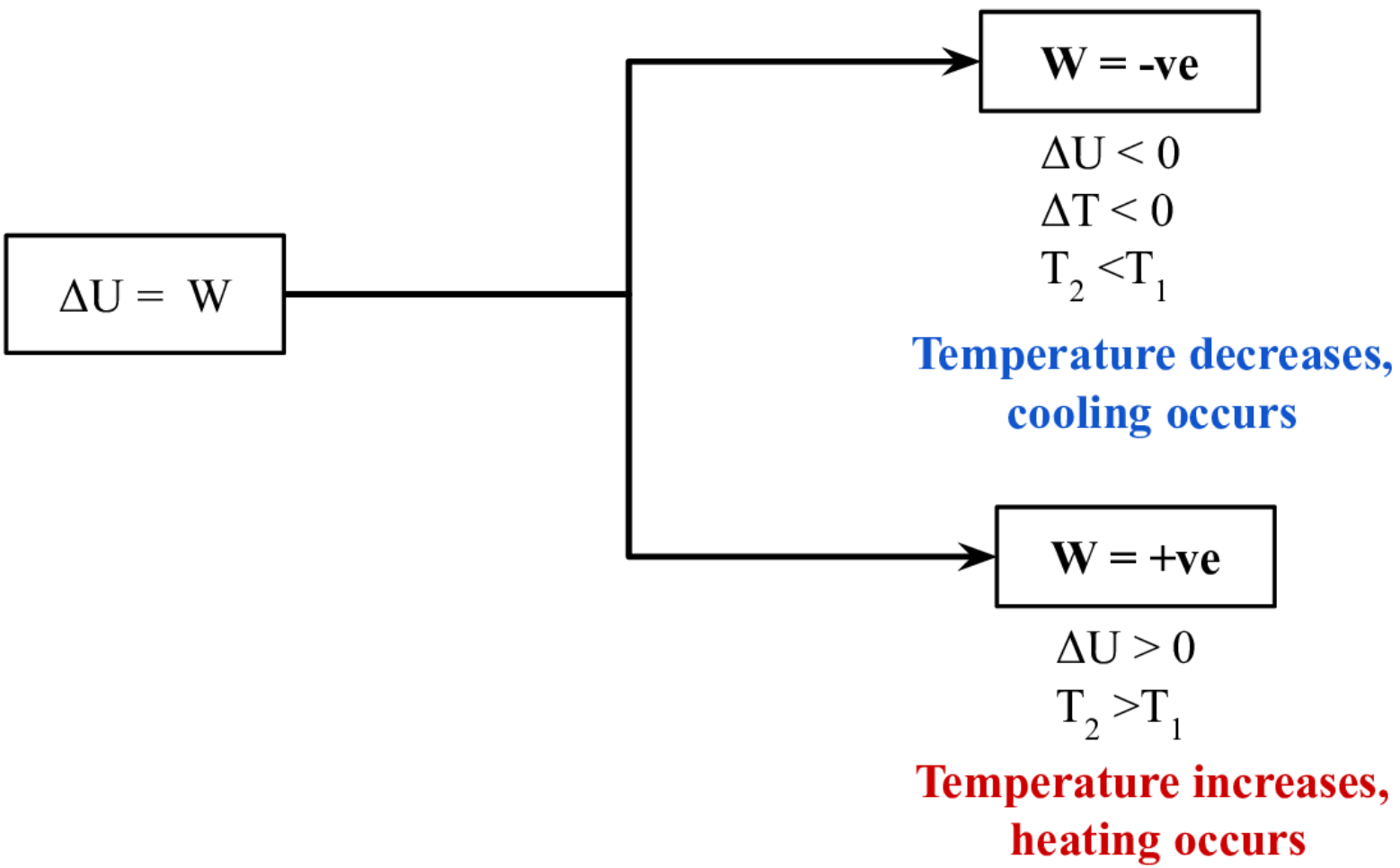
1. Relationship between Cp & Cv
2. **Calculation of Adiabatic Work**

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Work in Adiabatic Process

There is no heat exchange between system and surrounding, $q = 0$

From first law of thermodynamics, $\Delta U = q + W$



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Calculation of Adiabatic Work

Operation of a Adiabatic Reversible Process

$$dq = 0$$

$$dU = dW$$

$$nC_V dT = -PdV$$

$$\int nc_V dT = \int -\frac{nRT}{V} \cdot dV$$

$$\int_{T_1}^{T_2} \frac{C_V . dT}{T} = -\int_{V_1}^{V_2} \frac{R}{V} . dV$$

$$C_V \ln \frac{T_2}{T_1} = -R \ln \frac{V_2}{V_1}$$

$$\ln \frac{T_2}{T_1} = \ln \left(\frac{V_2}{V_1}\right)^{R/c_V}$$

$$\Rightarrow \left(\frac{T_2}{T_1}\right) = \left(\frac{V_2}{V_1}\right)^{-R/c_V}$$

$$T_2 V_2^{\gamma-1} = T_1 V_1^{\gamma-1}$$

or $$T V^{\gamma-1} = \text{constant}$$

$$PV^{\gamma} = \text{constant}$$

NOTE: These expressions are only valid for adiabatic reversible process.

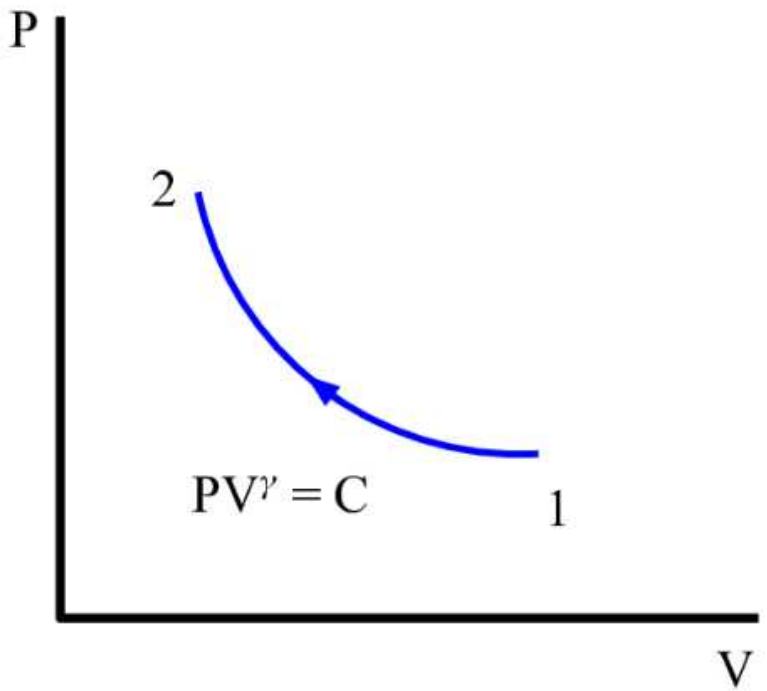
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Calculation of Adiabatic Work

Operation of a Adiabatic Reversible Process

Thus, for an **adiabatic reversible** process:

$$PV^{\gamma} = \text{constant}$$



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Adiabatic Ideal gas (Exp / Comp).

$$dU = dq + dw$$

$$dU = dw$$

$$dU = n C_v dT$$

$$dw = n C_v dT$$

($PV = nRT$, $T = \frac{PV}{nR}$)

$$= n \frac{R}{\gamma - 1} (T_2 - T_1).$$

$$= \frac{nR}{\gamma - 1} \left(\frac{P_2 V_2}{nR} - \frac{P_1 V_1}{nR} \right) = \frac{P_2 V_2 - P_1 V_1}{\gamma - 1}$$

$$W = \frac{P_2 V_2 - P_1 V_1}{\gamma - 1}$$

→ Reversible / Irreversible

16. 10 mole of ideal gas expand isothermally and reversibly from a pressure of 10 atm to 1 atm at 300 K. What is the largest mass which can lifted through a height of 100 meter?

(a) 31842 kg
(b) 58.55 kg
(c) 342.58 kg
(d) None of these

$$W = -2.303 n R T \log \frac{P_i}{P_f}$$

$$W = mgh$$

(@harithachem)

Calculation of Adiabatic Work

Work done in an Adiabatic Reversible Process

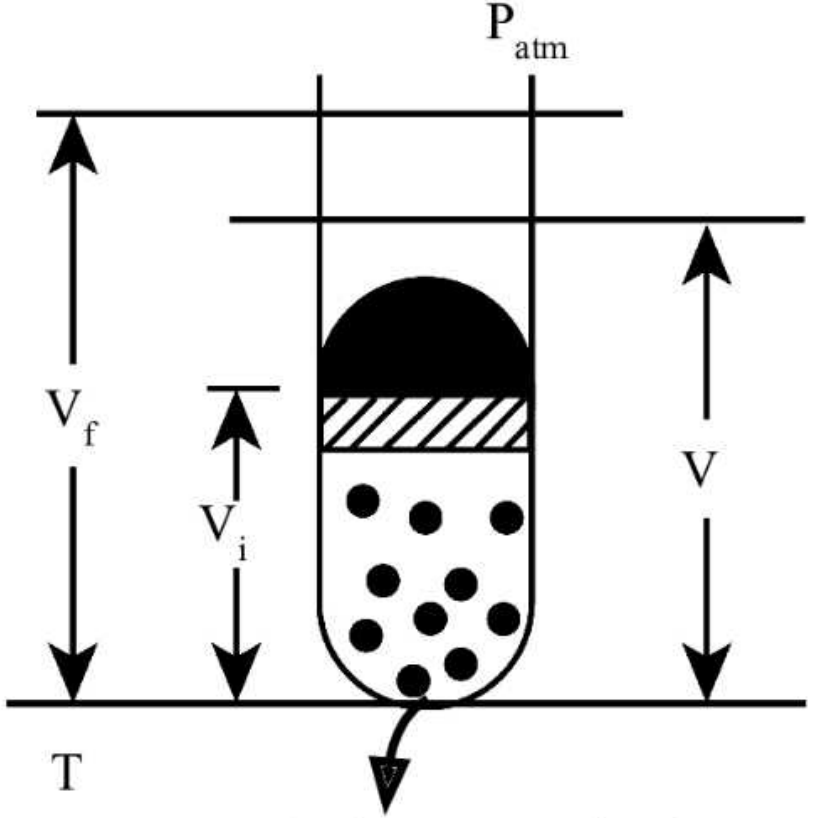
$$w = -\int P_{ext}.\, dv, \qquad \text{but } P_{ext} = P_{int} = \frac{K}{V^{\gamma}}$$

$$\therefore\ w = -\int \frac{K}{V^{Y}}\ dv$$

$$\Rightarrow w = -K\,[V_2^{\,1-\gamma} - V_1^{\,1-\gamma}]/(1-\gamma\}$$

$$= \frac{P_2V_2^{\gamma}.V_2^{\,1-\gamma} - P_1V_1^{\gamma}.V_1^{\,1-\gamma}}{\gamma-1}$$

$$\Rightarrow \text{work done} = \frac{P_2V_2 - P_1V_1}{\gamma-1}$$



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Calculation of Adiabatic Work

Work done in an Adiabatic Irreversible Process

The work done (w) is given by

$$\Delta U = w = -P_{ext}(V_f - V_i)$$

$$\Delta U = nC_V(T_f - T_i)$$

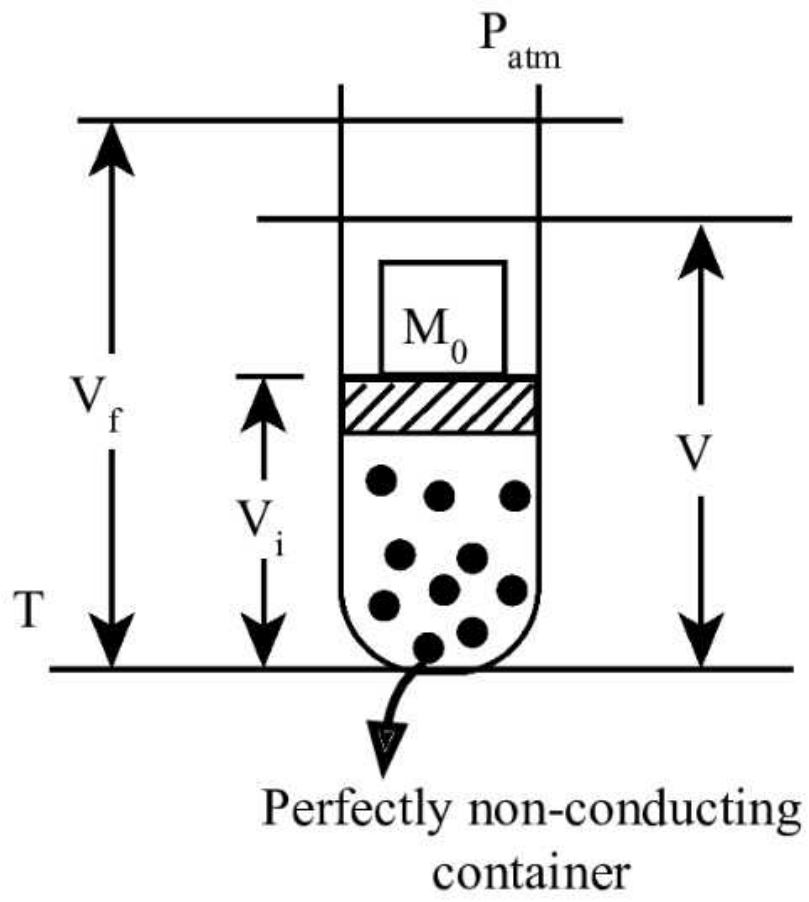
$$\Rightarrow nC_V(T_f - T_i) = -P_{ext}(V_f - V_i)$$

$$nC_V(T_f - T_i) = -P_{ext}\left(\frac{nRT_f}{P_f} - \frac{nRT_i}{P_i}\right)$$

Solving this equation for T_f

now $w = \Delta U = nC_V(T_2 - T_1)$

$$w = \left(\frac{P_fV_f - P_iV_i}{\gamma - 1}\right)$$



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Comparison of Isothermal and Adiabatic Process

Isothermal process

$PV = C$

$PdV + VdP = 0$

$VdP = -PdV$

$$\frac{dP}{dV} = -\frac{P}{V}$$

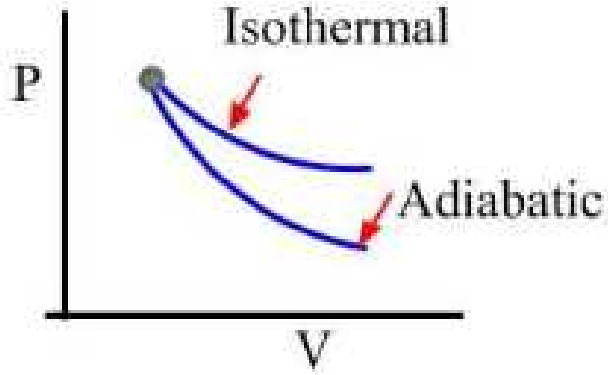
Adiabatic process

$PV^{\gamma} = C$

$P\,\gamma\,V^{\gamma-1}.dV + V^{\gamma}.dP = 0$

$V^{\gamma}\,dP = -P.\gamma.V^{\gamma}.V^{-1}.dV$

$$\frac{dP}{dV} = -\frac{P\gamma}{V}$$



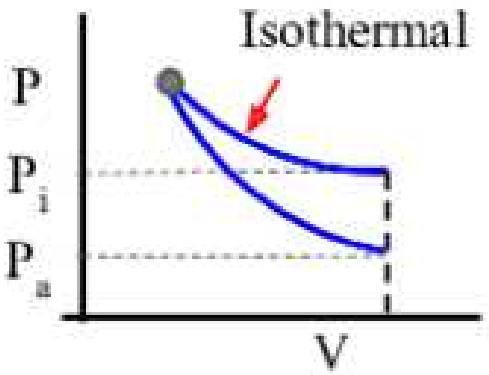
Adiabatic slope = γ × isothermal slope

Since γ is always >1, slope of P-V curve is more negative in case of adiabatic process.

Comparison of Isothermal and Adiabatic Process

Case - I: For same final volume

(a) Expansion:

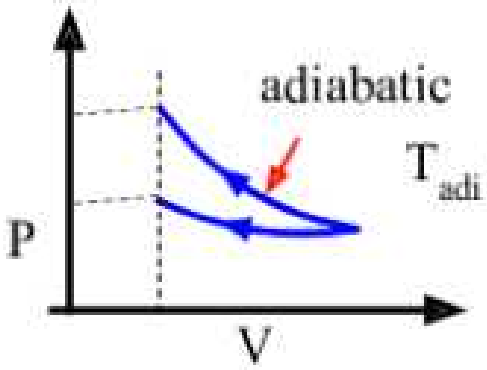


$P_i > P_a$

$\Rightarrow T_i > T_a$

$|w_{isothermal}| > |w_{adiabatic}|$

(b) Compression:



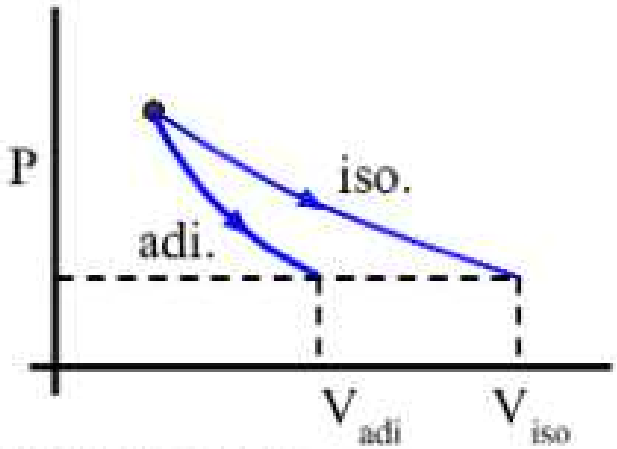
$T_{adi} > T_{iso}$

$|w_{adiabatic}| > |w_{isothermal}|$

Comparison of Isothermal and Adiabatic Process

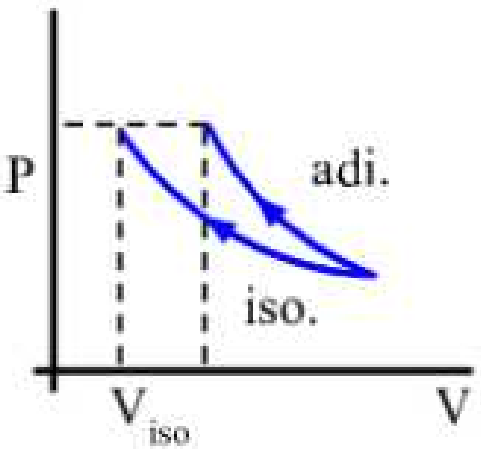
Case - II: For same final pressure

(a) Expansion:



$$|w_{isothermal}| > |w_{adiabatic}|$$

(b) Compression:



$$|w_{adiabatic}| > |w_{isothermal}|$$

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Interactivity Pod

PV = Constant
It is valid for isothermal as well as adiabatic reversible process.

A True

B False

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Interactivity Pod

PV = Constant
It is valid for isothermal as well as adiabatic reversible process.

A True

B False

For adiabatic reversible process,
PV^{γ} = constant

Where γ is known as Poisson's ratio.

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Let's Solve.



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Let's Solve.

Temperature of the system decreases in an

A. Adiabatic compression

B. Isothermal compression

C. Isothermal expansion

D. Adiabatic expansion

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Let's Solve.

Temperature of the system decreases in an

A. Adiabatic compression

B. Isothermal compression

C. Isothermal expansion

D. Adiabatic expansion

Solution

During adiabatic expansion, work is done at an expense of the internal energy. Thus, temperature decreases.

Silent Mode

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For an adiabatic reversible process, $nC_V dT = -PdV$. (T/F)



Discuss among yourselves

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Doubts Time



I will now be taking frequently asked doubts

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Doubts



- Why is the P-V curve is more negative in case of adiabatic process than that in isothermal process ?

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Quiz.





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Quiz.

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JEE Mains 2017

ΔU is equal to

A. Isochoric Work

B. Isobaric Work

C. Adiabatic Work

D. Isothermal Work

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Quiz.

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JEE Mains 2017

ΔU is equal to

A. Isochoric Work

B. Isobaric Work

C. Adiabatic Work

D. Isothermal Work

Solution

For adiabatic process, $q = 0$. Thus, $\Delta U = w$.

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Quiz.

Chat Off

A gas expands adiabatically at constant pressure such that $T = V^{-1/2}$. The value of $\gamma(C_{p,m}/C_{v,m})$ of the gas will be

A 1.30

B 1.50

C 1.70

D 2

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Quiz.

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A gas expands adiabatically at constant pressure such that $T = V^{-1/2}$. The value of $\gamma(C_{p,m}/C_{v,m})$ of the gas will be

A 1.30

B 1.50

C 1.70

D 2

Solution

$$T \propto \frac{1}{\sqrt{V}}, \quad TV^{1/2} = \text{constant}$$

For adiabatic process,

$$TV^{\gamma-1} = \text{constant}$$

$$\therefore \gamma - 1 = \frac{1}{2}, \gamma = \frac{3}{2}$$

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Quiz.

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The work done by the gas in reversible adiabatic expansion process is :

A $\frac{P_2V_2 - P_1V_1}{\gamma - 1}$

B $\frac{nR(T_1 - T_2)}{\gamma - 1}$

C $\frac{P_2V_2 - P_1V_1}{\gamma}$

D None of these

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Quiz.

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The work done by the gas in reversible adiabatic expansion process is :

A $\frac{P_2V_2 - P_1V_1}{\gamma - 1}$

B $\frac{nR(T_1 - T_2)}{\gamma - 1}$

C $\frac{P_2V_2 - P_1V_1}{\gamma}$

D None of these

Solution

$$\Delta U = w = \frac{P_2V_2 - P_1V_1}{\gamma - 1}$$

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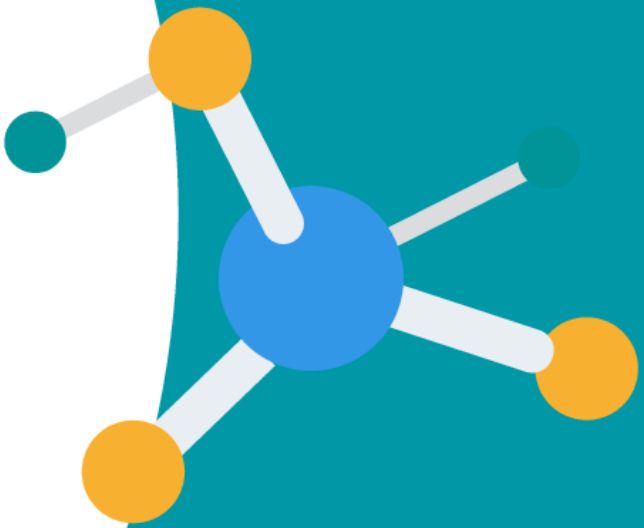
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Milestone Achieved

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Summary.



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Relationship between C_p and C_v

Table for C_P and C_V for different types of gases

Atomicity	C_V	$C_P = C_V + R$	$\gamma = \frac{C_P}{C_V}$
Monoatomic	$\frac{3}{2}R$	$\frac{5}{2}R$	$\gamma = \frac{5/2R}{3/2R} = 1.66$
Diatomic or linear polyatomic	$\frac{5}{2}R$	$\frac{7}{2}R$	$\gamma = \frac{7/2R}{5/2R} = 1.40$
Non - linear polyatomic	3R	4R	$\gamma = \frac{4R}{3R} = 1.33$

NOTE: These values are applicable at low or moderate temperatures.

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Summary

For an adiabatic process,

$\Delta U = w$

If work done by the system, T decreases.

If work done on the system, T increases.

Calculation of work for adiabatic process

$$W_{rev} = \frac{nRT_2 - nRT_1}{\gamma - 1} = \frac{P_2V_2 - P_1V_1}{\gamma - 1}$$

$$W_{irr} = \frac{nR(T_2 - T_1)}{\gamma - 1} = \frac{P_2V_2 - P_1V_1}{\gamma - 1}$$

Comparison of isothermal and adiabatic process

Slope: Adiabatic = $\gamma \times$ Isothermal

For expansion: $|w_{isothermal}| > |w_{adiabatic}|$

For compression: $|w_{adiabatic}| > |w_{isothermal}|$

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HOMEWORK

(Pick and Solve Based on Session Coverage)

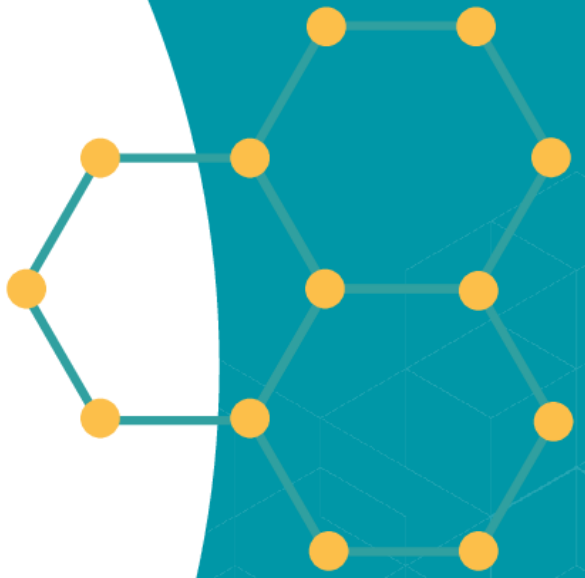
Topic name as per Curriculum tree	Assignment Detail	Tatva Exercise 1 question number
Introduction to thermodynamics	Assignment 1 (question 1, 3, 4, 12)	1, 2, 3, 4, 5, 6,
Types of processes	Assignment 1 (question 5, 6, 7, 8) Assignment 4 (question 13, 16, 17)	
Internal energy	Assignment 1 (question 9, 10, 13, 16) Assignment 2 (question 3, 9, 12)	7, 8, 9
Concept of Heat and work	Assignment 1 (question 2, 11, 14, 15) Assignment 2 (question 10)	10, 11, 12, 13
Zeroth law of thermodynamics	Assignment 2 (question 1)	
First law of thermodynamics	Assignment 2 (question 2, 7, 11, 13, 14)	14, 15, 16, 17
Reversible and irreversible work	Assignment 2 (question 4, 5, 6, 8, 15) Assignment 4 (question 18, 19)	18, 19, 20, 21, 22, 23, 24, 25, 26, 27
Heat Capacity of A System	Assignment 3 (question 1, 2, 3, 4, 12, 13) Assignment 4 (question 14)	28, 29
Adiabatic process and poisson's ratio	Assignment 3 (question 5, 6, 7, 8, 9, 10, 11, 14, 15, 16, 17)	30, 31, 32, 33, 34
Enthalpy of A System	Assignment 4 (question 6, 7, 8, 11, 12)	35, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48,
Gibb's Free Energy Change in Reaction	Assignment 6 (question 7, 14)	

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Feedback

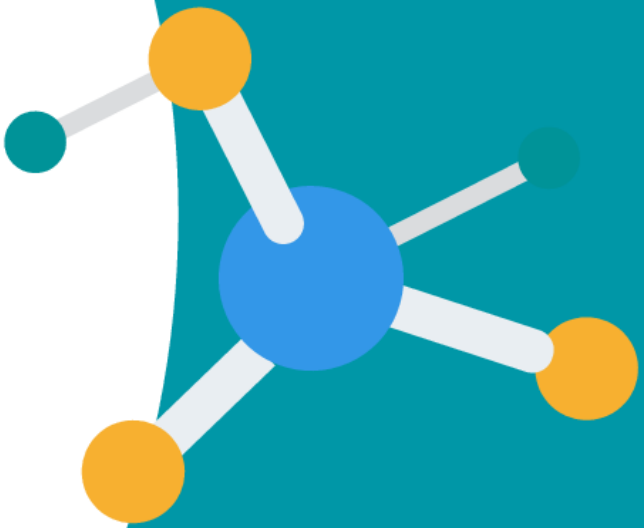


Thanks.



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Summary.



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Summary

Applications of first law

(a) For isobaric process:

$\Delta U = q - \int p.dV$

(b) For isothermal process:

$q = - w$ $\quad w = - p_{ex} \Delta V$ (irreversible) $\quad w = - 2.303nRT \log \frac{V_2}{V_1}$

(c) For cyclic process:

$q = - w;$ $\quad w =$ area under the P-V curve

(d) For isochoric process:

$\Delta U = q_v$

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Summary

Total Heat Capacity (C_T)

Heat required to raise the temperature of system by 1°C under the given process is known as total heat capacity.

Specific Heat Capacity (s)

Heat required to raise temperature of unit mass (generally 1 g) of a substance by 1°C.

Molar Heat Capacity (C_m)

Heat required to raise temperature of 1 mole of a substance by 1°C.

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HOMEWORK

(Pick and Solve Based on Session Coverage)

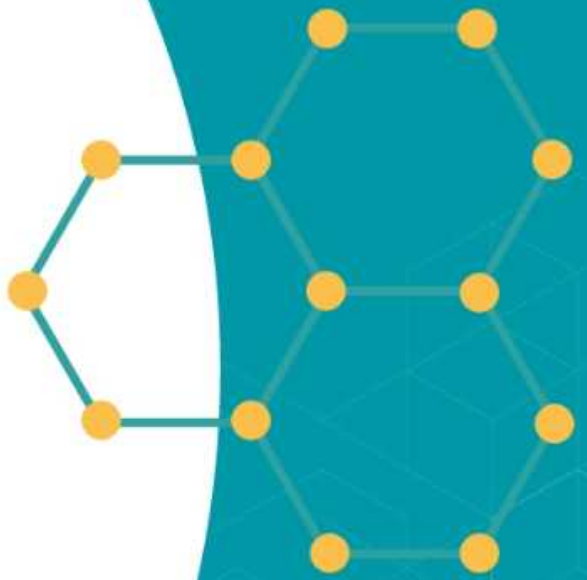
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Feedback



Thanks.



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